The Last Mile—2002

Richard P. Rumelt
2004

Jilted by broadband

By Michael V. Copeland and Om Malik
March 13, 2002

When Microsoft was developing the Xbox video game console in 1999, consumers were signing up for high-speed Internet access as quickly as they could. Seduced by this growth in broadband subscriptions, Microsoft made a critical decision. The device would be equipped with a broadband ethernet connection--instead of a conventional dial-up modem. . . . .Three years later, Microsoft's broadband-or-nothing decision looks premature. Only 8 percent of U.S. households--some 10.5 million--have a broadband connection, but 58 million households have a dial-up connection. And the rate of growth in the number of broadband subscriptions is flattening; it's been 15 to 20 percent per quarter over the last two quarters, compared with almost 100 percent per quarter in 1999. Before the Xbox release last fall, Microsoft's CEO, Steve Ballmer, tacitly acknowledged the looming problem to the European press. "We have grown more pessimistic about this than we were a few years ago," he said. "It [broadband] has proceeded more slowly than we expected."

CNNMoney, April 23, 2002 - Jupiter Media Metrix (NASDAQ: JMXI), the global leader in Internet and new technology analysis and measurement, today reports that while only 16 percent of U.S. online households subscribe to broadband, more than 24 percent of dial-up consumers are considering signing up for a broadband service within the next 12 months. Jupiter analysts have found that these dial-up consumers actually favor utility over entertainment, even though companies catering to a broadband audience emphasize entertainment and rich-media applications.

. . . one emerging trend is the focus, particularly among the DSL operators, to bundle access with content. For example, many of the Bell companies are stepping up alliances with AOL and Microsoft Network, which will give them prepackaged content. Qwest and MSN also are developing a special portal for Qwest DSL customers. Similarly, SBC and Yahoo! have announced plans to launch a co-branded broadband ISP by mid-2002.

ISP, 2/27/2002 by Roy Mark

Web surfers snub pricier broadband Internet access

04/25/2002 - Updated 11:24 PM ET
By Andrew Backover, USA TODAY

Broadband growth continues to slow as customers balk at higher prices and companies scale back promotions.

The number of new broadband subscribers grew 12.6% in the first quarter from the previous quarter, based on results from eight of 17 major players, research firm ARS says.

That means it's likely that the industry's overall growth will fall short of the 15.8% growth from the third to the fourth quarter last year.. . . . . Behind the trend:

- **Price.** The average monthly cost of cable modem service in the fourth quarter of last year rose to $44.22 — up 19% year-over-year. Monthly DSL prices rose 7% to $51.67. As consumers feel the economic slowdown, many are deciding to skip the extra expense. A recent survey by Jupiter Media Metrix found that 76% of dial-up users were not keen to pay more for broadband. While about 70 million U.S. households could get broadband, only 11.6 million do. Prices aren't likely to fall. Dozens of broadband players have folded or curbed expansion during the telecom implosion. "Pricing is the critical thing," says Dave Burstein of DSL Prime.
- **Marketing.** After rolling out big discounts late last year, which boosted growth rates for the first time in two years, many companies have now scaled back such deals because they hurt revenue, says TeleChoice analyst Pat Hurley. Late last year, Verizon offered DSL for the first three months at $29.95 monthly, plus a free camera. In March, the cost was $39.95 a month for the first three months, plus a gift. The $29.95-a-month offer resurfaced Thursday without the gift. Instead, customers get the first month free if they sign up online. Comcast was also offering free months last year, but no longer. "There is, in general, weak marketing by providers," Hurley says.
- **Content.** Though customers rave about faster speeds with broadband than with dial-up connections, compelling content, such as videos and music, is sparse. That does not "give people a reason to switch," Hurley says.

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By the end of the twentieth century, most of the core networks of the U.S. telephone system had been converted to high-capacity high-speed optical fiber, and almost all telephone switches were high-speed digital. However, the connections between the core network and the individual customer—called the "local loop" in the telecommunications industry—remained largely copper wire (called "twisted pairs" in the industry). With the rapidly rising interest in the Internet and the promise of e-commerce, video-on-demand, and other high-bandwidth applications, more and more consumers and business people saw the local loop as the bottleneck preventing widespread high-speed access. Professionals and college students were especially aware of the dramatic difference between Internet access at 1-10 Mbps (million bits per second), commonly available on shared Ethernets, and access at 40-46 Kbps (thousand bits per second) available on a standard "56 Kbps" analog modem connected to telephone lines. No matter how fast the core network became, consumer access to the Internet would be limited unless the problem of the "last mile," the limitation of the final connection between the networks and the consumer, could be overcome.

In 1999, the deployment of generally available high-speed access over the local loop had seemed to be a sure thing. Nearly everyone in the computer and telecommunications industry forecast that falling prices of bandwidth would combine with a public appetite for high-speed data services (e.g., streaming video, live video-conferencing) to accelerate broadband access in the last mile. However, by 2002 questions were surfacing about the whole enterprise. The key issues being discussed were (1) How "fast" is high-speed access? (2) Was there really a public demand for truly high-speed services? (3) Were the DSL and cable technologies being deployed poor substitutes for true fiber-based broadband? (4) In the present race, which technology had the edge, DSL or cable?

The primary technologies being deployed to solve the problem of the last mile were DSL service over existing copper lines and 2-way data service over upgraded cable TV systems. Two other technologies were being studied and deployed in very limited trails: wireless broadband and optical fiber to the home. As of early 2002, no one was sure how rapidly broadband would be adopted by consumers, or what ultimate market shares DSL and cable solutions would obtain. This note provides background information on each of these technologies and the costs associated with each.

Broadband

Electronic information is transmitted over channels that have a property called "bandwidth." The most elementary analog signals are variations of intensity at a fixed frequency. For example, middle C on the piano is a tone that comes from vibration at 440 cycles per second, or Hertz (Hz). The highest tones most people can hear are about 16 KHz (Kilo-Hertz), and the lowest are about 25 Hz. Thus the frequency "bandwidth" of the human ear is 16,000-25 = 15,075 Hz. To convey sonic information accurately, an analog electronic system would require a bandwidth of at least 16 KHz. Most of the information in the human voice, as another example, is carried by sounds with frequencies in the range 300 to 3,300 Hz. So the frequency bandwidth[1] of the normal telephone line was designed to be 3,000 Hz—adequate for communication but not sufficient for high-fidelity music.

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[1] The traditional communications definition of "bandwith" is in terms of a range of frequencies. In recent years, interest in communications has grown beyond communications engineers, and people have come to use the term bandwidth to mean the maximum data rate possible on a channel. To distinguish between these two very different meanings, we shall use the term "frequency bandwidth" to refer to the classical concept.

Similarly, the frequency bandwidth required by a standard broadcast (analog) television channel was 6 MHz (Mega-Hertz, or million cycles per second). Channel 2, for example, was broadcast on a frequency 57 MHz and "modulated" with the television information. The resulting electronic signals had frequencies ranging from 54-60 Mhz. Channel 3 resided in the spectrum 60-66MHz, and so on.

The relationship between bandwidth and information capacity follows from mathematical information theory, developed by Shannon [1948]. His key theorem states that the capacity *C* of a channel in (bits/second) is related to bandwidth B (in Hz) by the formula:

$$C = B \log_2 \left(1 + S / N\right),$$

where S/N is the signal to noise ratio. Shannon's theories led to the development of highly sophisticated methods of coding and decoding information to extract the maximum bit-rate, or information rate, out of a channel. For example, a typical telephone line has a signal-to-noise ratio of 1000. Given a frequency bandwidth of 3000 Hz we have C = 3000 $\log_2(1001)$ = 30 Kbps. That is, if the true frequency bandwidth of a telephone line is 3000 Hz, the best possible modem can only provide a data rate of 30 Kbs over that line (Hence the 28.8 Kbs modem). Faster modem speeds have been obtained by exploiting the fact that most telephone lines have bandwidths above 3000 KHz.

One TV channel has a frequency bandwidth of 6 Mhz, and a signal-to-noise ratio of about 100, providing a maximum channel capacity of 40 Mbps. Whereas that is the channel capacity of the bandwidth granted to an analog TV signal, the information in a digital TV broadcast can be compressed (MPEG), or coded, to require only 10 Mbps. The frequency bandwidth required to transmit 10 Mbps is only 1.5 MHz. So by moving to digital representation and encoding, four times as many TV channels can be transmitted in the same frequency bandwidth.

Until the late 1990s, 2-way channels of communication into residences and small businesses were narrowband (low bit rate) voice channels, carried either over the local loop or a wireless link. Television, by contrast, was a high bandwidth 1-way channel. The promise of *broadband access* was that high-bandwidth 2-way communications channels could be provided on a mass basis.

In some sense, the issue was simply one of cost. The standard data line offered by telcos was called a T1. A T1 line provided a steady 1.5 Mbps of capacity, which was more than sufficient for most people's web surfing. However, telco pricing for a T1 was about $500 for installation, $500/month for the line, and an additional $500/month for an ISP to link it to the Internet. A T3 connection, providing 45 Mbps, could cost more than $5,000 per month. Looking into the far future, technologists imagined that each residence and business would be directly connected to the core network by optical fiber channels. In 2002, one optical strand of fiber was capable of data rates up to 2.5 Gbps with standard equipment, and 100 Gbps with sophisticated equipment. The latter capacity was sufficient to carry (simultaneously) all of the long-distance voice traffic in the United States.[2] Researchers believed that, ultimately, the capacity of each fiber strand could be increased by a factor of 100.

The broadband technologies being deployed in 2002 were each compromises between bandwidth and cost. But each promised to provide millions of residences with access to the Internet at speeds from 250Kbps to 6 Mbps at reasonable prices.

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[2] The data rates achieved over a fiber optic link depended upon the attached photo-electronic equipment and the network architecture feeding it.

Core Bandwidth

Huge investments in fiber-optic backbone plant connecting cities and continents had been made in the period 1995-99. The major long distance carriers (AT&T, MCI WorldCom, and Sprint) and the specialized fiber carriers (Qwest, Global Crossing, Level 3, and Williams) were each spending billions of dollars to build out national networks. A national fiber backbone connected about 110-150 cities with fiber cables containing 48-96 fiber strands. Ordinarily, the builder would "light" only 4-8 of the fibers; the remaining "dark fiber" was left for future growth. Some companies sold interest in the dark fiber to finance the project, dramatically reducing the cost to them of the "lit" fiber. Eight fiber strands connecting Los Angeles to San Francisco could carry 640 Gbps of traffic. And each of six or eight companies might have ten times this amount in reserve. The result was what analysts were calling a "glut" in long-distance fiber capacity.

One effect of the glut was dropping long-distance telephone charges. In the period 1984-1997, fourteen years following the de-regulation of AT&T and the introduction of competition into the long-distance market, the price of long-distance voice telephony had remained largely unchanged. Dropping local access charges reduced the costs of calling coast-to-coast, but the long-distance component had barely fallen. Then, in the period 1998-2000, prices began to decline steeply. In part this was due to the initiation of single-rate pricing (e.g., 10 cents/minute), in part it was due to RBOCs[3] entering the long-distance market, and in part it was due to the glut of fiber capacity to carry increased traffic.

The expectation lying behind much of the fiber that had been placed was that the Internet would launch a growing wave of data traffic. Although data traffic was growing at a rapid pace, there was a very sharp contrast between the availability of enormous bandwidth in the backbone and the limitations of the last mile.

Demand

In 2002 there were 107 million households in the U.S., about 55% having a PC. Of the PC-enabled households, it was estimated that 90% or more had some form of online Internet access. Of the online households, approximately 20% had a high-speed connection.

Looking to the future, households were expected to grow at about 1% per year; PC penetration was expected to stabilize at about 60% of households. It was expected that all household PCs would be online by 2004. The great uncertainty lay in the proportion that would utilize broadband high-speed services. Some analysts put the figure as high as 50% of online households by 2005, whereas others believed that mass broadband access would "always" be 3-5 years away, and saw the broadband market as limited to perhaps one-third of online users.

Exhibits 1-4 show basic data about PC usage, Internet usage, and broadband usage among U.S. households in late 2001. Exhibit 5 shows a forecast prepared by McKinsey and JPMorgan. Exhibit 6 shows actual broandband subscriber counts by the survey firm TR Online.

Another uncertainty lay in the degree to which the "set-top-box" provided a broadband Internet channel for households which did not have or want a PC. The "set-top-box" was an upgraded cable TV box which handled (a) digital cable TV, (2) high-speed Internet access with TV screen display, and (3) voice telephone services. Some industry observers thought that the future belonged to this technology and

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[3] Deregulation of AT&T resulted in it being split into the long-distance carrier, still called AT&T, and seven Regional Bell Operating Companies, or RBOCS.

that the cable companies would emerge as the power players in the new media. Others, like Apple's Steve Jobs, believed that TV and the Internet were two different media and two different mental states:[4] "When you watch TV you sit back and turn your brain off. When you work with your computer you sit up close to the screen and turn your brain on. They are totally different media."

Pricing in the high-speed market was another source of uncertainty. The initial DSL offerings in 1999 tended to be priced at about $45 per month. Since then, prices had risen by about $10 per month. Cable prices had been stable at about $45 per month. At this price the customer could buy a 256 Kbps DSL connection or a cable modem connection that ran at higher speeds as long as not too many neighbors were also using the system.

In a 2001 study, McKinsey and JPMorgan identified several waves of potential residential broadband buyers. The early adopters had already bought (11% of online homes). An additional 10% of online homes expressed "high interest" and were expected to adopt in the near future. Most interesting was the fact that fully 17% of non-early adopter online homes expressed an interest in broadband if it were $40 per month or less. Furthermore, these same online homes were ***already*** paying about $38 per month for a dedicated telephone line plus an Internet ISP. Were they to switch to DSL service on their primary telephone line, they could save the $38.

DSL

DSL stood for "digital subscriber line." The promise of DSL technology was that it used existing copper connections into residences to supply data rates up to 6 Mbps. DSL technology came in various flavors (HDSL, ADSL, SDSL, VDSL, etc.), each representing a different method of encoding data and a different set of limitations on its use. In general, the higher bandwidth DSL technologies required "cleaner" copper plant and shorter runs between the customer and the telephone central office (or remote terminal). DSL technology was not new: it had been in use within telcos for years as a method for managing communications across the system.

The version of DSL discussed in this note was normally termed ADSL; the "A" stood for "asymmetric," a reference to the fact that the upstream data rate was normally less than the downstream data rate.

DSL service was provided over a regular telephone line, which retained its usability for regular telephone service. The trick was using frequencies above the range of human hearing to carry digital information. The advantage was that existing copper wires could be used. The disadvantage was that the line had to be relatively short (less than 12-18,000 feet), have no connectors in its run, and not suffer interference from other close-by DSL lines.

When a customer requested DSL service, the telco first checked the line to see if the service was possible. In some cases, the line had to be conditioned by field technicians, who removed old filters and re-arranged connections so that the line could be used for DSL signals. It cost an average of $500 to condition a line. Telco executives estimated that about 10 percent of standard copper lines needed such conditioning in order to provide DSL service.

To receive DSL service, the user needed a "splitter". This was a small device that "split" the incoming line into two separate telephone lines: one carrying ordinary voice frequencies and the other carrying the special high-frequency data information.

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[4] Interview, March 1998.

Without a splitter, telephones would be damaged and the DSL modem would not perform correctly. If the residence had, or could accommodate, a telephone circuit that could be set aside for DSL service, the splitter was installed in the terminal box where the telephone line connected to the house. The average cost to a telco of having a splitter installed was $85.

Recently, telephone companies had begun to encourage DSL subscribers to "self-install" by using micro-splitters. Under this approach, the combined voice and data signals were distributed throughout the residence on existing lines, and small splitters were placed at each telephone jack: one set to pass data traffic and all the others set to pass only voice information. By 2002, about two-thirds of new DSL installations were self-installs, so that the average installation cost per new subscriber had fallen to $90·(1/3)= $30.

In addition to the splitter, the telco normally supplied the user with a DSL modem. This device converted the high-frequency information into a standard Ethernet data stream. An external DSL modem plugged into a PC Ethernet port, or an internal DSL modem plugged into a motherboard slot. The average cost of a modem and associated micro-splitters was $160.

Once the connection was established, the installer or user pointed the operating system's TCP/IP control logic to the appropriate servers, and access was established.

In a classic DSL system, the user's copper line ran directly to the telco's central office (CO). There it was tied to a DSLAM (DSL Access Multiplexer). The DSLAM combined (multiplexed) the data signals from numerous DSL lines into large capacity data trunks. It also passed the voice traffic over to the regular public switched network.

A DSLAM cost $250-500 per line. The DSLAM was, in turn, attached to a DS1 (1.5 Mbps) or DS3 (45 Mbps) data link to the Internet. Wholesale prices on a DS3 were about $5-8000 per month, depending upon the distance to a primary Internet node.

Whereas a standard DS3 data line was warranted to carry a steady flow of 45 Mbps, 24 hours a day, DSL service depended upon the "bursty" and non-steady demands created by most Internet users. Statistically, a user with a 512Kbps DSL connection used this maximum data transfer rate only about 5% of the time. That meant that the combined needs of 20 such users approximated a steady flow of 512 Kbps. Overloading by 20 meant that a DS3 line could handle 1800 DSL users. At a more conservative overloading of 10, it could handle 900 users (at 512Kbps each).

Industry experts estimated that about 44% of U.S. telephone lines were DSL addressable in 1999. Ironically, these lines were not in the newest neighborhoods. Over the years 1980-1999, many telcos had responded to the growing demand for second lines and new lines in new neighborhoods by installing small sub-stations (steel boxes next to the street), called DLCs (digital loop carriers), which carried high-speed digital network links closer to customers. The DLCs compressed and combined the voice traffic from about 100 homes into digital signals for carriage to and from the central office; often using a T1 link for the backhaul to the central office. The problem with this architecture was it left no extra capacity for DSL service. The upgrades telcos had provided to the newer neighborhoods blocked the easy roll-out of DSL.

The cost of upgrading old DLCs depended upon the density of the area served. Putting in fiber, rather than T1, links and tying the system together cost an average of $200 per line passed—that is, for each telephone line in the area, whether or not it was a DSL line. As shown in Exhibit 3, the cost of upgrade in the least dense zone (D) was 70% higher than in the most dense zone (A). Exhibit 5 provides data on the distribution of U.S. households both by neighborhood density level and by household

income. Once a DLC system was upgraded, it still required an additional investment to supply each DSL subscriber: mini-DSLAM cards had to be inserted into the DLC cabinets at a cost of about $250 per DSL subscriber

In 1999, only 44% of telephone lines were DSL addressable (closer than 15,000 feet to a CO and/or not served by an old-style DLC). By early 2002, 70% of lines were DSL addressable. More precisely, of the 200 million telephone lines in the U.S., 63% were directly connected to a CO, and 37% were connected to a DLC. Of those directly connected to a CO, about 11% were too far for DSL service, and 22% were unusable for other reasons, leaving 42% of all lines directly addressable from a central office. Of those attached to DLCs, about 25% were attached to old-style DLCs, and 75% were attached to new-style or upgraded DLCs.

The telco would bear several non-recurring costs when adding a new DSL subscriber. First, it was estimated that the cost of acquiring a new data customer was about $125. This average figure included marketing, promotion, initial customer contacts, and contracting. In addition, as noted above, the line might have to be conditioned, DSLAM cards installed in the CO or the DLC, splitters installed, and a DSL modem provided.

Recurring expenses included customer service, billing, network maintenance, and Internet access expenses. Residential customer service was estimated to require 1.5 telephone conversations per month at the start, each call costing $5. This cost was expected to decline to fall by 50% every 18 months for a continuing customer.

Billing and other administrative expenses ran about $4.50 per subscriber per month. If operations could be fully standardized and rolled into the telco’s standard billing operation, there would be little if any billing expense. However, changing rate structures, partnerships, and service packages had kept that from happening to date.

Network maintenance was expected to cost about $2.75 per month per subscriber and to remain constant over time.

Internet access costs varied with the type of connection and with the telco's contracts with other carriers. On average, 512Kbps DSL service cost about $10 per month per subscriber.

All of these costs are summarized in Exhibit 12.

HFC Cable Connections

The telephone network had been configured from the start as a 2-way communications system. By contrast, cable TV systems had been designed for one-way traffic only—they carried TV signals from a head-end station to customers' TV sets. The typical head-end station served 10,000 subscribers and provided a cable connection with a 350 Mhz bandwidth. This bandwidth enabled the system to carry about 50 channels of regular analog television. The subscriber used a "set-top-box" to tune into one of the 50 channels.

In 2000, about 60% of households subscribed to a cable TV service. Cable systems passed almost everyone; only 7% of households were not passed by a cable TV service. To upgrade cable services meant increasing the bandwidth carried and making provision for 2-way communications. On average, this upgrade would cost about $235 per home passed, with very wide variations depending upon density. The costs of cable plant upgrades for four density levels are given in Exhibit 4.

Cable companies had looked at the possibility of offering digital TV and some 2-way capabilities (including voice traffic) over their systems since the early 1990s. But for most, the capital costs seemed prohibitive. But the unexpected success of direct broadcast satellite television (e.g., DirectTV) pushed the industry towards a re-

sponse. During 1997-2001 the major cable operators (Cox, AT&T, Time Warner, Cablevision, Adelphia, and Comcast) had spent many billions upgrading their cable plant. The standard upgrade extended the bandwidth from 350 (or 500) Mhz up to 750 Mhz and added 2-way capabilities. The upgraded systems were termed "HFC," with the acronym standing for Hybrid Fiber Cable. By 2002, HFC systems passed 80 million households, with build-out plans in place to extend that reach to 90+ million households by 2004. Upgraded cable passed about 80% of the 20 million "high-income" households in the U.S.

HFC systems had about 3,750 Mbps of downstream capacity and 75 Mbps of upstream data capacity. However, television channels normally took about 90% of the downstream capacity. The remaining capacity was shared among the users on a local *node*. The typical node provided 500 homes with 27 Mbs of downstream capacity and 10 Mbps of upstream capacity. If the 5% usage pattern held, 500 households could share this capacity and each would experience about 1 Mbps of downstream capacity. Upstream capacity was much more fragile. Silicon Valley cable companies had quickly found that they could not support customers who tried to operate web sites on the cable system! With 250 homes on a feeder line, users would experience 2-4 Mbps of upstream capacity at a 5% penetration (12.5 homes) level. If 25% of homes were data subscribers (62.5 homes), upstream capacity per household would drop to 400-800 Kbps. And at 50% penetration levels, upstream capacity fell to 200-400 Kbps.

Issues with limited upstream capacity were of little concern as long as users stuck to web-browsing. However, sustained file transfers, even if pure downloads, strained the system. This occurred because of the structure of the Internet's TCP/IP system for packet transmission. According to this protocol, each group of data sent down stream had to be acknowledged by an upstream ACK packet before the next set of downstream data could be sent. If the ACK was not received, the sender asked again; if the ACK was received late, the sender slowed down the file transmission. Consequently, users who were all downloading files were also continually sending out streams of ACK packets over the upstream link. If that link congested, ACK packets went out late for everyone, slowing down everyone's downloads.

The average non-recurring costs associated with the addition of a new data subscriber were $470. This included the costs of a cable modem, 1 truck rolls at $100, $125 for customer acquisition, and $125 for network equipment (the counterpart to DSLAMS in telco systems). Exhibit 13 provides the details of these costs.

Monthly recurring expenses were roughly comparable to those of a telco. Network maintenance was slightly less, and Internet access had traditionally cost slightly more. Until recently, cable companies contracted with ISPs to handle their subscriber's Internet data and e-mail, the largest such ISP being Excite@Home. However, at the end of 2001, Excite@Home had gone bankrupt, cutting off service to its customers. Cable companies scrambled to replace the ISP and generally found that they could do so at the prices ($12 per month) it had been charging, or less. Of course, they could not easily replace its home pages and brand name, but there was a questions as to the value of those assets.

Voice on Cable

Once cable systems had been upgraded to handle 2-way communications, there was the opportunity to offer voice services. It was reported that this had been AT&T's motive in its purchase of TCI. By using that firm's extensive cable operations, AT&T was able to offer data and voice services to customers, breaking through the local loop that otherwise insulated it from its customers.

By early 2002, AT&T had established that it could offer voice over cable, but it had also found that quality of service was a problem. Even though customers had become used to “best-efforts” service in mobile applications, they appeared to continue to expect guaranteed telephony from hard-wired systems.

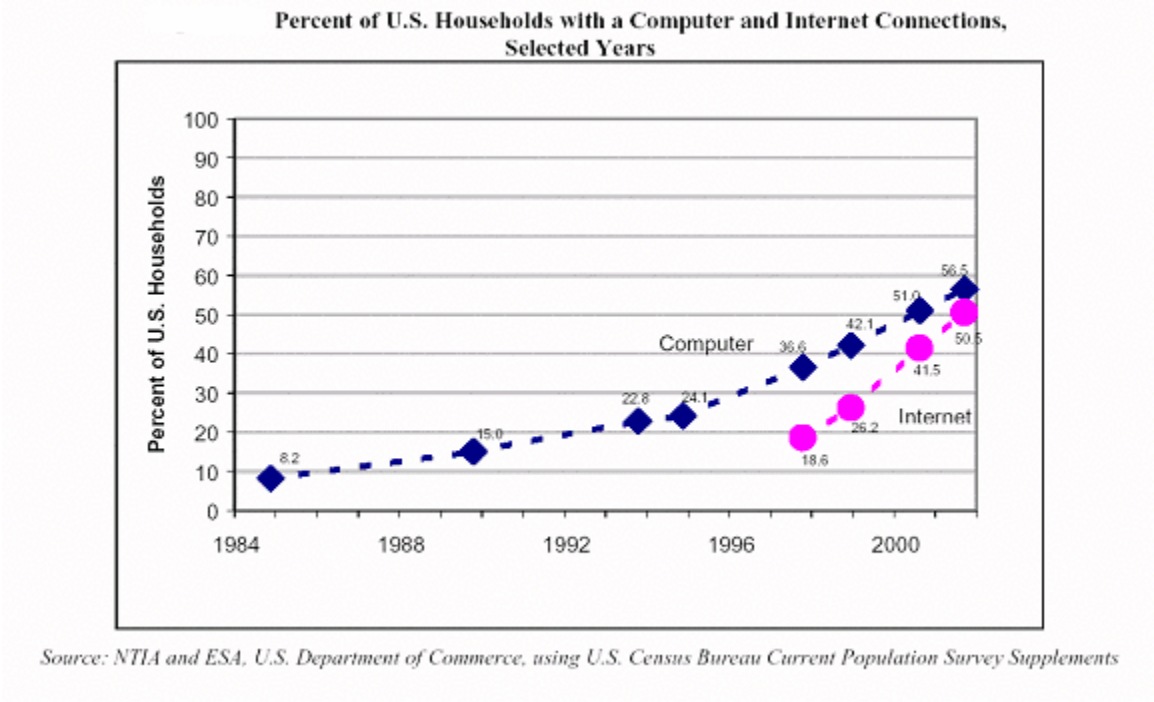
VOIP was less expensive than circuit-switch telephone for three reasons: the equipment used was simpler, the organizations creating the service did not have legacy infrastructures and overheads to support, and data traffic was not tariffed in the same way as voice traffic. For example, when a long distance carrier (e.g., MCI/Worldcom) delivered a voice call to a local telephone company for further delivery to the customer, it paid the local telco an access charge. If, however, the traffic was carried as Internet data, no such access charge was payable. The long distance carrier could, in principle, convert the data to an analog signal in facilities within the destination region and complete the transaction with a fixed tariff local call.

To handle voice traffic, a cable company would have to "harden" its system (improving the signal-to-noise ratio). This would cost about $25 per home passed. The costs of adding each new voice subscriber would be about $370 (see Exhibit 9 for details). This cost included an integrated set-top-box, which would handle digital and analog TV, high-speed data, and voice. It also included $50 for a battery—telephone customers were used to the idea that the telephone network had its own power and worked even when the electricity was off. A cable telephony business would have to either put batteries in each set-to-box or provide power over the cable itself. The latter alternative would require capital investments of about $20 per home passed.

Many industry observers thought that a cable company offering television, data, and voice services would be a strong competitor to telcos. The customer of such a company could literally cut off his telco wire and immediately payback the cost of his TV service.

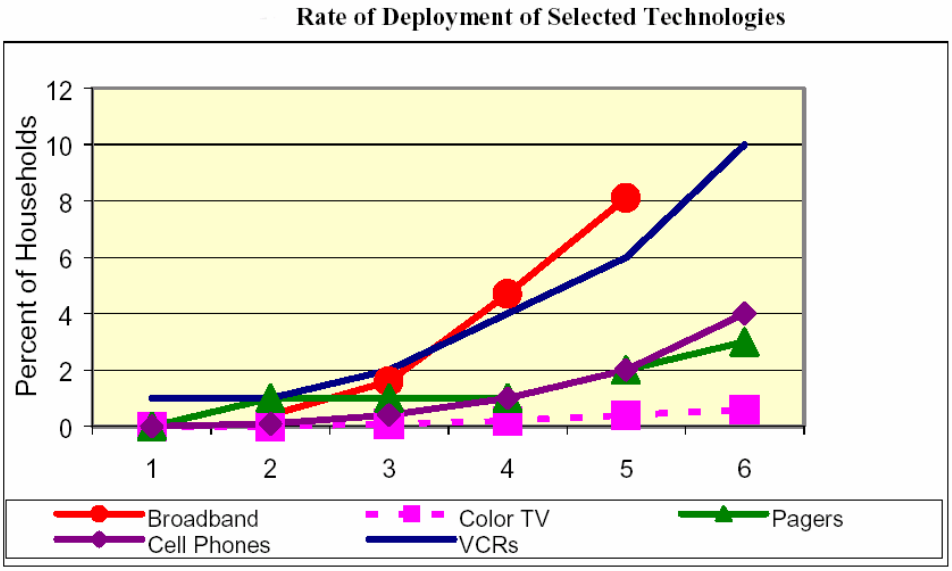
There were also opportunities for telcos to offer video services to their customers. VDSL technology allowed the customer to have voice, high-speed data, and high-quality digital video delivered over copper pairs. The catch was that the system only worked on line lengths of 5000 feet or less and required fiber networks laid to each node serving 100 homes. The capital investments involved in this kind of network upgrade made many telco executive look at the alternative of fiber direct to the home. A fiber "pipe" directly connecting each household to the backbone was the dream of every technologist—the information superhighway brought to each home. The bandwidth available with a fiber connection would permit new services such as full video on demand, easy videoconferencing, and was a precondition to full virtual reality systems whose value and impact were only dimly understood. But fiber to the home looked to cost about at least $5000 per home. Costing close to $500 billion, rewiring the United States with fiber to the home seemed a project only the government could undertake. Some analysts, however, expected these costs to fall in half in five years, so that the full fiber network would eventually be built.

Exhibit 1



Source: NTIA and ESA, U.S. Department of Commerce, using U.S. Census Bureau Current Population Survey Supplements

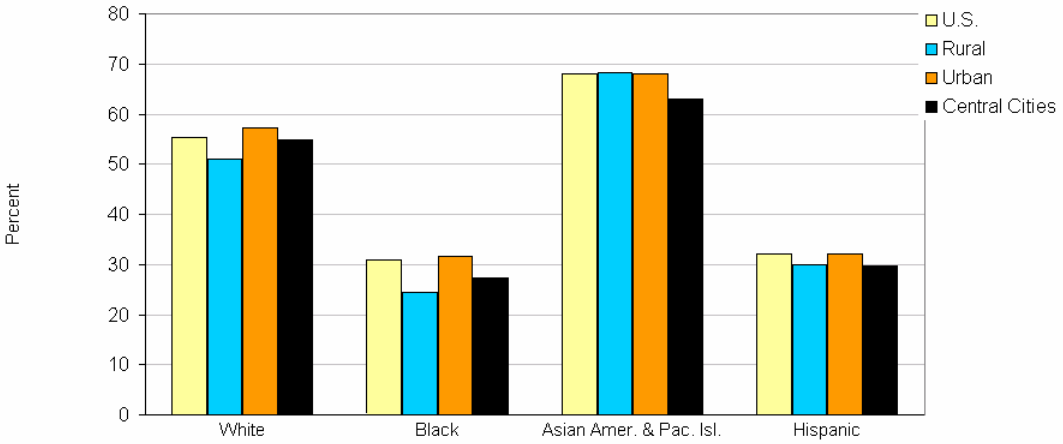
Exhibit 2



Source: eBrain Market Research and the National Cable and Telecommunications Association

Exhibit 3

Percent of U.S. Households with Internet Access, By Race/Hispanic Origin, By U.S., Rural, Urban, and Central Cities, 2001

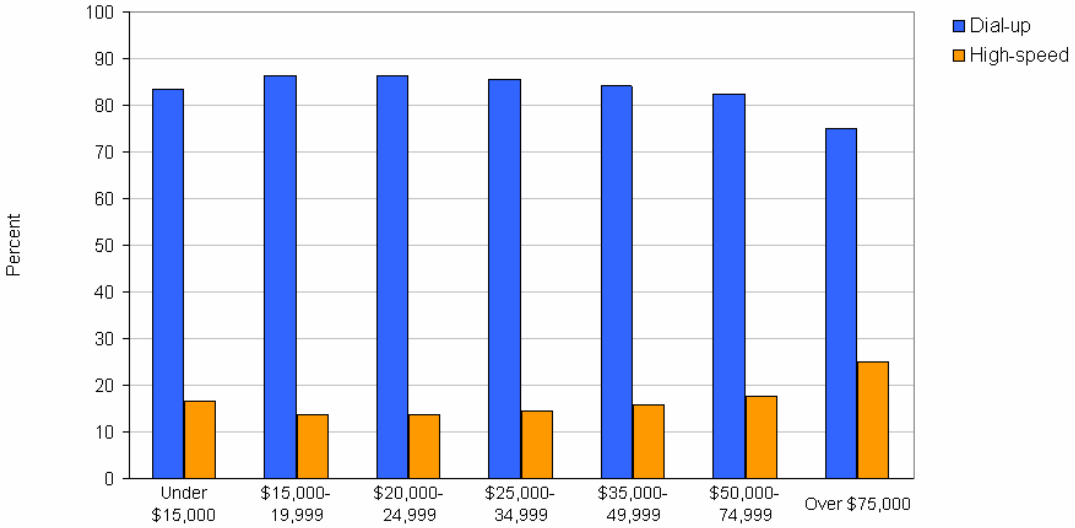


	White	Black	Asian Amer. & Pac. Isl.	Hispanic
U.S.	55.4	30.8	68.1	32.0
Rural	51.0	24.4	68.2	29.9
Urban	57.2	31.6	68.1	32.2
Central Cities	54.8	27.4	63.1	29.8

Source: NTIA and ESA, U.S. Department of Commerce, using U.S. Bureau of the Census Current Population Survey supplements.

Exhibit 4

Percent of U.S. Households with Internet Access, By Access Speed, By Income, 2001



	Under $15,000	$15,000-19,999	$20,000-24,999	$25,000-34,999	$35,000-49,999	$50,000-74,999	Over $75,000
Dial-up	83.4	86.4	86.2	85.4	84.1	82.4	74.9
High-speed	16.7	13.6	13.8	14.6	15.9	17.7	25.1

Source: NTIA and ESA, U.S. Department of Commerce, using U.S. Bureau of the Census Current Population Survey supplements.

Exhibit 5: McKinsey & JPMorgan Forecast

			Thousands			
	1999	2000	2001E	2002E	2003E	2004E
US Households	104	105	106	108	109	110
PC Penetration	48%	53%	53%	55%	55%	57%
PC Households	49.9	55.7	56.4	59.2	60.0	62.9
Online Penetration	86%	88%	90%	94%	97%	100%
Households Online	42.9	49.0	50.8	55.7	58.2	62.9
Broadband Penetration	4%	11%	22%	30%	38%	48%
Boradband Households	1.7	5.4	11.2	16.7	22.1	30.2

Exhibit 6: TR Online Broadband Subscriber Census (millions of subscribers)

	4Q-99	1Q-00	2Q-00	3Q-00	4Q-00	1Q-01	2Q-01	3Q-01	4Q-01	1Q-02
Cable	1.58	2.28	2.8	3.51	4.18	4.93	4.93	5.86	6.63	7.69
DSL	0.07	0.19	0.29	1.27	2.36	2.40	3.10	3.10	3.97	4.39
Total	1.65	2.47	3.09	4.77	6.54	7.33	8.03	8.96	10.59	12.08
Growth		50%	25%	54%	37%	12%	9%	12%	18%	14%

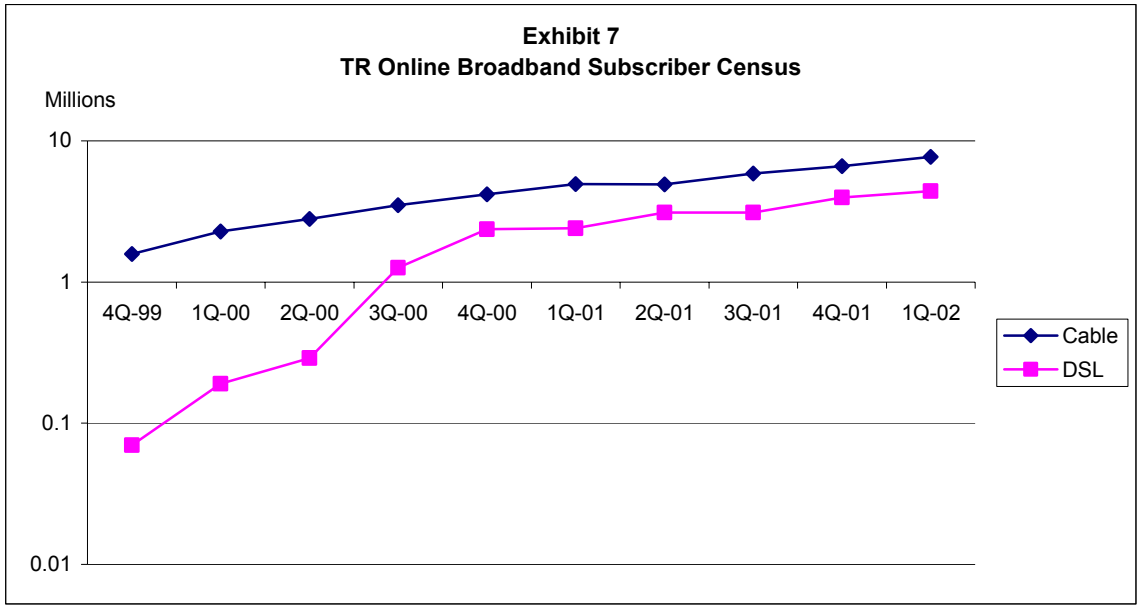


Exhibit 8: Capital Cost of Telco-DLC Upgrades by Density

HP = Homes Passed

Density Class	Density HP/Mile	HP (Millions)	Percent HP	Capital Cost per HP ($)	% Lines on DLC	% DLC's Old Tech.
A	145	45	42%	175	30%	27%
B	73	36	34%	200	40%	28%
C	45	18	18%	230	45%	20%
D	25	6	6%	300	50%	22%
All	96	105	33%	200	37%	26%

Exhibit 9: Capital Cost of Cable System Upgrades by Density

Density Class	Density HP/Mile	HPs (Millions)	Percent HPs	Capital Cost per HP ($)
A	145	40	42%	125
B	73	32	34%	235
C	45	17	18%	350
D	25	6	6%	700
All	95	95	33%	239

Exhibit 10: Distribution of U.S. Households by Income and Regional Density

Density Zone	Quintiles of Household Income (1=lowest, 5=highest) 1	2	3	4	5	Total
A	7.1%	7.2%	8.4%	9.8%	9.6%	42.1%
B	6.1%	6.2%	6.8%	7.0%	8.3%	34.4%
C	4.0%	4.1%	3.6%	2.2%	1.6%	15.5%
D	2.8%	2.5%	1.2%	1.0%	0.5%	8.0%
Total	20.0%	20.0%	20.0%	20.0%	20.0%	100.0%

Exhibit 11: Average Household Income by Quintile and Density Zone

$ Thousands/Year

Density Zone	Quintiles of Household Income (1=lowest, 5=highest) 1	2	3	4	5	Overall Average
A	9	21	39	60	117	54
B	12	25	41	64	140	62
C	8	23	36	55	122	37
D	7	20	32	52	90	26
Overall Average	9	23	39	60	126	

Exhibit 12: Cost of Upgrading Telco Network, Installing and Operating DSL Servic

Telco DSL Economics	Cost	Est. Life
System investment to upgrade old DLC equipment		10
A density zone	175.00	15
B density zone	200.00	15
C density zone	230.00	15
D density zone	300.00	15
Non-recurring costs of adding a data subscriber		
Customer acquisition expense	125.00	6
Installation expense	30.00	10
Customer DSL modem	160.00	8
CO Service		
Loop qualification	50.00	20
DSLAM	350.00	8
CO service total non-recurring	715.00	
DLC Service		
Mini-DSLAM card in DLC	250.00	8
DLC Service non-recurring	565.00	
Recurring expense of a data subscriber ($/month)		
Customer service	8.00	
Administration	4.50	
Network maintenance	2.75	
Internet access	10.00	
	25.25	

Exhibit 13: Cost of Upgrading Cable Systems, Installing and Operating Data Services

HFC Cable System Data Economics	Cost	Est. Life
Capital costs to upgrade from 350 mhz to 750 mhz 2-way cable		
A density zone	125.00	15
B density zone	235.00	15
C density zone	350.00	15
D density zone	700.00	15
Non-recurring costs of adding a data subscriber		
Customer acquisition expense	125.00	6
Installation expense	100.00	10
Customer cable modem	120.00	8
Head-end data overlay equipment	125.00	8
Total non-recurring	470.00	
Recurring expense of data subscriber		
Customer service	7.00	
Administration	2.00	
Network maintenance	2.00	
ISP fee	12.00	
	23.00	

Exhibit 14: Costs of Upgrading HFC for Voice Services

HFC Telephony	Cost
Capital expense of hardening the system	25.00
Set-top box	280.00
VOIP network equipment	120.00
Installation	120.00
Battery	50.00
Cost of transport and access	7.00
Billing and customer service	3.50